

ASSIGNMENT 4

Hashing

Learning outcome:

1. Applying solution using various hashing methods.
2. Implementation of the hashing.

Dateline: 24 January 2026, upload to PutraBLAST

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1. Consider inserting the keys 10, 22, 31, 4, 15, 28, 17, 88, and 59 into a hash table of length $m=11$ using open addressing with the primary hash function:

$$h_1(k) = k \bmod m$$

Illustrate the result of inserting these keys using linear probing and using double hashing with $h_2(k) = 1 + (k \bmod (m-1))$.

2. Write a program to insert sequence of items to hash table using hash function, $h_1(k) = 2x \bmod m$, where k is the keys and m is the size of the hash table. Use linear probing policy for any collision. Then, find an item in the hash table. Shows the index of the item that you are looking for, otherwise, write an error message.

Example: $m = 9$, $k = \{45, 23, 102, 96, 116, 87\}$

0	1	2	3	4	5	6	7	8
45	23		96	87		102	116	

Operation find 87: output – index 4

Operation find 100: output - error message “*item not found*”